

Channel representations of noisy coherent detection in continuous-variable quantum key distribution: Additive-noise versus attenuator

A. S. Naumchik,^{1,*} Roman Goncharov,^{1,2,†} and Alexei D. Kiselev^{3,4,‡}

¹*Quantum Information Laboratory, ITMO University,
Kadetskaya Line, 3/2, St. Petersburg, 199034, Russia*

²*SMARTS-Quanttelecom LLC, 6th Line of Vasilyevsky Island, 59/1B, St. Petersburg, 199178, Russia*

³*Laboratory for Quantum Communications, ITMO University,
Birzhevaya Line, 16, Saint Petersburg 199034, Russia*

⁴*Laboratory of Quantum Processes and Measurements,
ITMO University, Kadetskaya Line 3b, Saint Petersburg 199034, Russia*

(Dated: August 14, 2026)

We study the security of continuous-variable quantum key distribution (CV-QKD) under receiver asymmetry within an untrusted-detector model. We analyze additive-noise and attenuator channel representations. For homodyne and double-homodyne detection, we derive the corresponding channel parameters and compare the resulting Eve–Bob Holevo information and asymptotic secret fraction.

For the parameter sets considered, receiver asymmetry reduces the asymptotic secret fraction in both channel representations. The representation giving the lower asymptotic secret fraction depends on the receiver parameters and the transmittance of the communication channel. For homodyne detection, the attenuator representation gives a lower asymptotic secret fraction at high transmittance and remains positive up to a longer distance.

In the double-homodyne case, the channel parameters, Eve–Bob Holevo information, and asymptotic secret fraction depend on the squeezing parameter r that labels the reference Gaussian POVM used in the channel representation. For the parameter sets considered, the Eve–Bob Holevo information as a function of r is concave for the additive-noise representation and convex for the attenuator representation. The choice of r and the channel representation therefore affect the resulting asymptotic secret fraction.

I. INTRODUCTION

Homodyne and double-homodyne detection are standard coherent detection schemes in quantum optics and continuous-variable quantum information [1, 2]. Here double-homodyne detection denotes the two-branch receiver that implements the conventional heterodyne measurement. In the strong local oscillator limit, an ideal homodyne receiver implements a projective measurement of one field quadrature. An ideal balanced double-homodyne receiver jointly estimates two conjugate quadratures through a coherent-state positive operator-valued measure (POVM), whose effects are proportional to coherent-state projectors [2, 3]. Both schemes implement Gaussian POVMs and give Gaussian outcome distributions for Gaussian input states. They are used in quantum state tomography [4, 5], quantum state preparation [6], and continuous-variable quantum key distribution (CV-QKD) [7–9].

Arthurs and Kelly gave an early quantitative treatment of joint measurements and derived an uncertainty relation for unbiased simultaneous measurements of conjugate observables [10]. Holevo represented a joint measurement of two canonical quadratures using an auxiliary mode in a fixed state and two commuting quadrature combinations [11]. He subsequently showed that, under a total mean-square-error criterion for an isotropic Gaussian state, the minimum is attained by a coherent-state POVM realized with a vacuum auxiliary mode [12]. Ali and Emch modeled a finite-resolution position measurement by averaging the corresponding projective measurement over a classical resolution kernel, which yields a POVM [13]. Yuen and Chan later analyzed quantum noise and excess noise from the local oscillator in optical homodyne and heterodyne detection [14]. Holevo’s modern classification includes single-quadrature homodyne measurements and joint two-quadrature heterodyne measurements; both classes allow additional Gaussian measurement noise [3].

Practical receivers may have detector quantum efficiencies below unity and beam-splitter imbalance. Within the Gaussian approximation to the photocount statistics in the strong local oscillator limit, these imperfections yield noisy Gaussian POVMs [15, 16]. These receiver imperfections enter CV-QKD parameter estimation and the calculation of

* Email address: naumchik95@gmail.com

† Email address: rkgoncharov@itmo.ru

‡ Email address: alexei.d.kiselev@gmail.com

the asymptotic secret fraction [17–19]. We use *receiver asymmetry* for the combined effect of beam-splitter imbalance and unequal detector quantum efficiencies. For each detection scheme, the receiver parameters determine the noisy Gaussian POVM to be reproduced, which we call the target POVM. We represent the target POVM by an effective one-mode Gaussian channel $\mathcal{C}$ acting after the communication channel and before a reference Gaussian POVM.

We restrict the analysis to zero-displacement Gaussian channels with centered noise and use units in which the vacuum covariance matrix is $\mathbb{1}$. If $\mathbf{r}_G$ and Σ_G are the first-moment vector and covariance matrix of a Gaussian state, we define both channels through their action on these moments:

$$\mathbf{r}_G \mapsto X_{\mathcal{C}} \mathbf{r}_G, \quad \Sigma_G \mapsto X_{\mathcal{C}} \Sigma_G X_{\mathcal{C}}^T + Y_{\mathcal{C}}.$$

For measurement type $\gamma \in \{\text{H}, \text{DH}\}$, the additive-noise channel Φ_{γ} and the attenuator channel $\mathcal{E}_{\gamma}$ are specified by

$$\begin{aligned} X_{\Phi}^{(\gamma)} &= \mathbb{1}, & Y_{\Phi}^{(\gamma)} &= \Sigma_N^{(\gamma)} \succeq 0, \\ X_{\mathcal{E}}^{(\gamma)} &= \cos \theta_{\gamma} \mathbb{1}, & Y_{\mathcal{E}}^{(\gamma)} &= \sin^2 \theta_{\gamma} \Sigma_{\text{env}}^{(\gamma)}, \end{aligned}$$

where H and DH label homodyne and double-homodyne detection, $\Sigma_N^{(\gamma)}$ is the receiver excess-noise covariance matrix in the additive-noise channel representation, $\cos^2 \theta_{\gamma}$ is the attenuator transmissivity with $0 \leq \theta_{\gamma} < \pi/2$, and $\Sigma_{\text{env}}^{(\gamma)}$ is the covariance matrix of a centered Gaussian state of the environmental input mode [2, 20]. For $\mathcal{C}_{\gamma} \in \{\Phi_{\gamma}, \mathcal{E}_{\gamma}\}$, the dual map $\mathcal{C}_{\gamma}^*$ is defined by $\text{Tr}[\mathcal{C}_{\gamma}(\hat{\rho})\hat{\Pi}] = \text{Tr}[\hat{\rho}\mathcal{C}_{\gamma}^*(\hat{\Pi})]$ for every state $\hat{\rho}$ and POVM effect $\hat{\Pi}$. We choose the channel parameters so that $\mathcal{C}_{\gamma}^*$, together with an invertible deterministic rescaling of the outcomes, maps the reference Gaussian POVM to the target POVM. The two channel representations therefore give the same receiver outcome statistics in the rescaled coordinates for every input state.

Let $\hat{\rho}_{\text{AB}}$ denote the state shared by Alice and Bob after the communication channel and before the effective receiver channel. Each channel representation produces

$$\hat{\rho}_{\text{AB}}^{(\mathcal{C}_{\gamma})} = (\mathcal{I}_A \otimes \mathcal{C}_{\gamma})(\hat{\rho}_{\text{AB}}),$$

where $\mathcal{I}_A$ is the identity channel on Alice's mode and $\mathcal{C}_{\gamma}$ acts on Bob's mode. Because the outcome rescaling is invertible, both representations give the same Alice–Bob mutual information. In a trusted-detector model, an eavesdropper (Eve) has no access to the environmental output of the effective receiver channel [21–24]. For a fixed joint state of Eve, Alice, and Bob before the effective receiver channel, the target POVM gives the same joint state of Eve and Bob's classical outcome, and hence the same Eve–Bob Holevo information, for both channel representations. In the untrusted-detector model adopted here, Eve holds a purification of $\hat{\rho}_{\text{AB}}^{(\mathcal{C}_{\gamma})}$. The two channel representations generally produce different states $\hat{\rho}_{\text{AB}}^{(\mathcal{C}_{\gamma})}$ and can give different values of the Eve–Bob Holevo information and asymptotic secret fraction. Selecting a channel representation therefore fixes an additional assumption about the effective quantum map used to represent receiver asymmetry.

Our previous work derived the target POVMs for receiver asymmetry in homodyne and double-homodyne detection and constructed their additive-noise channel representations [25]. For double-homodyne detection, we used a family of rank-one reference Gaussian POVMs generated by displaced squeezed states. The squeezing parameter r labels this family; $r = 0$ gives the coherent-state POVM of conventional double-homodyne detection. Positivity of the receiver excess-noise covariance matrix restricts r to an interval fixed by the receiver parameters. In the additive-noise channel representation, the Eve–Bob Holevo information depends on r under the untrusted-detector assumption. The target POVM admits several channel representations, each consisting of an effective channel, a rank-one reference Gaussian POVM, and an invertible outcome rescaling. Its outcome statistics alone do not determine $\hat{\rho}_{\text{AB}}^{(\mathcal{C}_{\gamma})}$ or the corresponding Eve–Bob Holevo information.

Here we construct an attenuator channel representation of the target homodyne and double-homodyne POVMs and compare it with the additive-noise channel representation. For homodyne detection, the environmental input state is the vacuum. For double-homodyne detection, the environmental input state is a centered pure Gaussian state. Its squeezing is finite in the interior of an admissible interval with nonzero width. At the interval endpoints, the attenuator representation is defined by limits with divergent environmental squeezing. If the interval collapses to a single value, the target POVM is rank one and both channel representations reduce to the identity channel. We derive the parameters of both representations by requiring the dual map and the outcome rescaling to reproduce the target POVM. We compare the resulting Eve–Bob Holevo information and asymptotic secret fraction for the Gaussian-modulated coherent-state protocol with reverse reconciliation under collective Gaussian attacks [7, 26, 27].

Section II defines the target homodyne and double-homodyne POVMs. Section III constructs their additive-noise and attenuator channel representations. Section IV evaluates the Alice–Bob mutual information, Eve–Bob Holevo information, and asymptotic secret fraction for both channel representations. Section V discusses the results.

II. TARGET GAUSSIAN POVMS

We now specify the target homodyne and double-homodyne POVMs produced by receiver asymmetry, as shown in Fig. 1. Receiver asymmetry arises from beam-splitter imbalance and unequal detector quantum efficiencies. Reference [25] gives the detailed derivations used in this section.

The Q symbol of the target homodyne POVM (see Fig. 1a) is

$$P_G^{(H)}(x) = \frac{1}{\sqrt{2\pi}\sigma_x} \exp\left[-\frac{(x - \langle \hat{x}_\phi \rangle)^2}{2\sigma_x}\right], \quad (1)$$

$$\langle \hat{x}_\phi \rangle \equiv \langle \alpha | \hat{x}_\phi | \alpha \rangle = 2 \operatorname{Re} \alpha e^{-i\phi}, \quad \phi = \arg \alpha_L, \quad (2)$$

$$\hat{x}_\phi = \hat{a}e^{-i\phi} + \hat{a}^\dagger e^{i\phi} \quad (3)$$

where α and α_L are the complex amplitudes of the signal and local oscillator (LO) modes, respectively, $\hat{x}_\phi$ is the phase-rotated quadrature operator, and x is the quadrature variable

$$x \equiv \frac{\mu}{(\eta_1 + \eta_2)CS|\alpha_L|} - \frac{\eta_1 S^2 - \eta_2 C^2}{(\eta_1 + \eta_2)CS} |\alpha_L| \quad (4)$$

and σ_x is the quadrature variance

$$\sigma_x \equiv \frac{\eta_1 S^2 + \eta_2 C^2}{[(\eta_1 + \eta_2)CS]^2}. \quad (5)$$

Taking the inverse Fourier transform of the Q symbol (1) gives the antinormally ordered characteristic function [28]. Its covariance matrix yields the target and reference homodyne covariance matrices

$$\Sigma_m^{(H)} = \lim_{r \rightarrow +\infty} R(\phi) \operatorname{diag}(e^{-2r} + \sigma_N, e^{2r}) R^T(\phi), \quad (6)$$

$$\Sigma_{\text{id}}^{(H)} = \lim_{r \rightarrow +\infty} R(\phi) \operatorname{diag}(e^{-2r}, e^{2r}) R^T(\phi), \quad (7)$$

where $\Sigma_{\text{id}}^{(H)}$ is the covariance matrix of the reference homodyne POVM, $R(\phi) = \begin{pmatrix} \cos \phi & -\sin \phi \\ \sin \phi & \cos \phi \end{pmatrix}$ is the rotation matrix, and σ_N is the receiver excess-noise variance,

$$\sigma_N = \sigma_x - 1. \quad (8)$$

For double-homodyne detection, the Q symbol of the target POVM is

$$P_G^{(\text{DH})}(x_1, x_2) = \frac{1}{2\pi\sqrt{\sigma_G^{(1)}\sigma_G^{(2)}}} \exp\left[-\frac{(x_1 - \operatorname{Re} \alpha e^{-i\phi})^2}{\sigma_1} - \frac{(x_2 - \operatorname{Im} \alpha e^{-i\phi})^2}{\sigma_2}\right], \quad (9)$$

$$\sigma_G^{(i)} = (\eta_1^{(i)} S_i^2 + \eta_2^{(i)} C_i^2) |\alpha_L^{(i)}|^2, \quad \alpha_L^{(1)} = C_L \alpha_L, \quad \alpha_L^{(2)} = -i S_L \alpha_L, \quad (10)$$

where $\sigma_G^{(i)}$ is the variance of the photon count difference before rescaling and x_i are the quadrature variables given by

$$x_1 = \frac{1}{2(\eta_1^{(1)} + \eta_2^{(1)})C_1 S_1 C_S} \left\{ \frac{\mu_1}{|\alpha_L^{(1)}|} - (\eta_1^{(1)} S_1^2 - \eta_2^{(1)} C_1^2) |\alpha_L^{(1)}| \right\}, \quad (11)$$

$$x_2 = \frac{1}{2(\eta_1^{(2)} + \eta_2^{(2)})C_2 S_2 S_S} \left\{ \frac{\mu_2}{|\alpha_L^{(2)}|} - (\eta_1^{(2)} S_2^2 - \eta_2^{(2)} C_2^2) |\alpha_L^{(2)}| \right\}, \quad (12)$$

and relations

$$\sigma_1 = \frac{\sigma_x^{(1)}}{2C_S^2}, \quad \sigma_2 = \frac{\sigma_x^{(2)}}{2S_S^2}, \quad (13)$$

$$\sigma_x^{(i)} = \frac{\eta_1^{(i)} S_i^2 + \eta_2^{(i)} C_i^2}{[(\eta_1^{(i)} + \eta_2^{(i)})C_i S_i]^2} \quad (14)$$

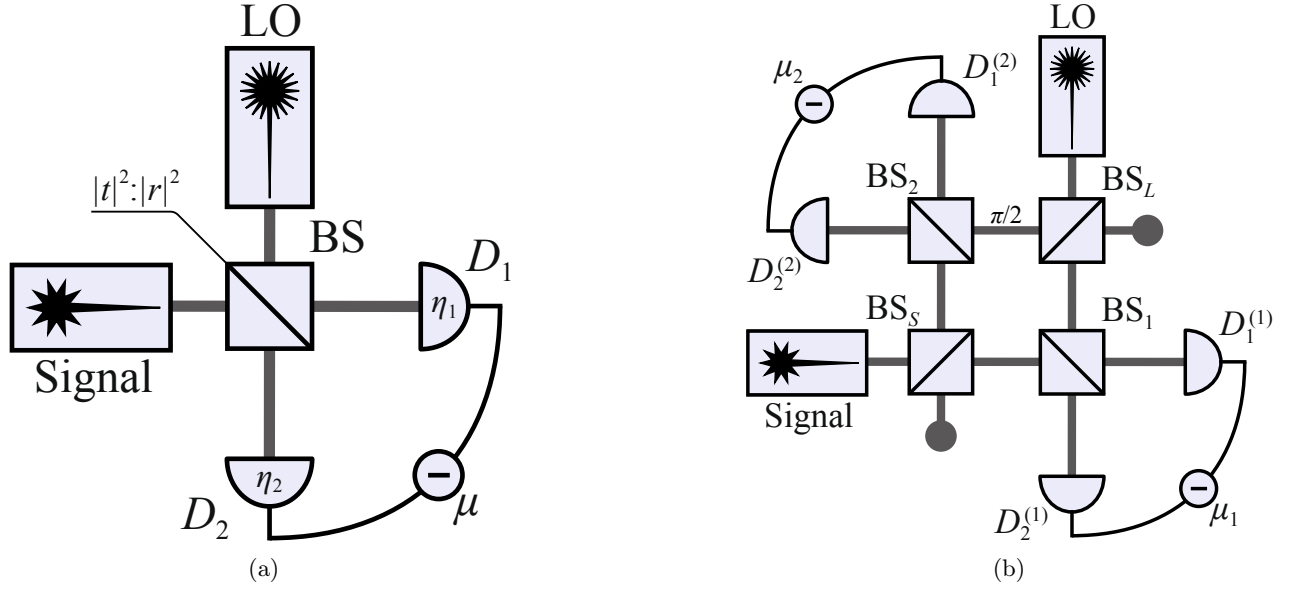


Figure 1: (a) Scheme of homodyne receiver. S (LO) is the source of signal (reference) mode, BS is the beam splitter with transmission and reflection coefficients C and S , respectively; photodetectors D_1 and D_2 have quantum efficiencies η_1 and η_2 , and $\mu = m_1 - m_2$ is the photon count difference. (b) Scheme of double-homodyne receiver. S (LO) is the source of signal (reference) mode, BS_S (BS_L) is the signal mode (local oscillator) beam splitter; $\pi/2$ is the quarter wave phase shifter; BS_i is the beam splitter of i th homodyne; $D_{1,2}^{(i)}$ are the photodetectors of the i th homodyne; and $\mu_i = m_1^{(i)} - m_2^{(i)}$ is the photon count difference registered by the detectors of i th homodyne.

give the quadrature variances σ_1 and σ_2 .

Following the same procedure used to derive Eq. (6), we obtain the covariance matrix of the target double-homodyne POVM,

$$\Sigma_m^{(\text{DH})} = R(\phi) \text{diag}(\delta_1, \delta_2) R^T(\phi), \quad (15)$$

$$\delta_i = 2\sigma_i - 1. \quad (16)$$

The channel representation uses a rank-one reference Gaussian POVM whose effects are proportional to projectors onto the displaced squeezed states

$$|z, r, \phi\rangle = \hat{R}(\phi) \hat{D}(z) \hat{S}(r) |0\rangle = \hat{D}(ze^{i\phi}) \hat{S}(re^{2i\phi}) |0\rangle, \quad (17)$$

where z is the displacement amplitude, r is a real squeezing parameter, and ϕ is the squeezing phase. At $r = 0$, the reference Gaussian POVM reduces to the coherent-state POVM implemented by conventional ideal double-homodyne detection. Here $\hat{D}(z) = e^{z\hat{a}^\dagger - z^*\hat{a}}$ is the displacement operator and $\hat{S}(r) = e^{\frac{1}{2}(r\hat{a}^{\dagger 2} - r^*\hat{a}^2)}$ is the squeezing operator.

The covariance matrix of the reference state is

$$\Sigma_{\text{id}}^{(\text{DH})} = R(\phi) \begin{pmatrix} e^{2r} & 0 \\ 0 & e^{-2r} \end{pmatrix} R^T(\phi). \quad (18)$$

The difference between the covariance matrices of the target POVM and the reference Gaussian POVM must be positive semidefinite. Equations (15) and (18) therefore restrict r to the interval fixed by the receiver parameters [25],

$$r \in [r_2, r_1], \quad r_1 = \ln \delta_1^{1/2}, \quad r_2 = \ln \delta_2^{-1/2}. \quad (19)$$

III. GAUSSIAN CHANNEL REPRESENTATIONS OF THE TARGET POVMS

For arbitrary state $\hat{\rho}$ and POVM $\hat{\Pi}^{(x)}$, the outcome probability is [29]

$$p(x|\hat{\rho}) = \text{Tr} \left(\hat{\rho} \hat{\Pi}^{(x)} \right). \quad (20)$$

We represent the target POVM by applying a channel $\mathcal{C}$ to the state before a reference Gaussian POVM $\hat{\Pi}_{\text{id}}$. Equivalently, the dual map $\mathcal{C}^*$ acts on the reference Gaussian POVM [20, 28]:

$$p(x|\hat{\rho}_G) = \text{Tr} \left\{ \mathcal{C}(\hat{\rho}_G) \hat{\Pi}_{\text{id}}^{(x)} \right\} = \text{Tr} \left\{ \hat{\rho}_G \mathcal{C}^* \left(\hat{\Pi}_{\text{id}}^{(x)} \right) \right\}. \quad (21)$$

Distinct Gaussian channels can induce the same target POVM after an invertible deterministic rescaling of the outcomes. This equality requires the same covariance matrix, first moments, and POVM normalization in the rescaled outcome coordinates.

We restrict $\mathcal{C}$ to Gaussian channels. Such a channel is characterized by matrices $X_{\mathcal{C}}$ and $Y_{\mathcal{C}}$. It transforms the first-moment vector $\mathbf{r}_G$ and covariance matrix Σ_G of a Gaussian state as [2]

$$\begin{aligned} \mathbf{r}_G &\mapsto X_{\mathcal{C}} \mathbf{r}_G, \\ \Sigma_G &\mapsto X_{\mathcal{C}} \Sigma_G X_{\mathcal{C}}^T + Y_{\mathcal{C}}. \end{aligned} \quad (22)$$

For invertible $X_{\mathcal{C}}$, the dual action on the parameters of a Gaussian POVM, expressed in the rescaled outcome coordinates, is

$$\begin{aligned} \mathbf{r} &\mapsto X_{\mathcal{C}}^{-1} \mathbf{r}, \\ \Sigma &\mapsto X_{\mathcal{C}}^{-1} (\Sigma + Y_{\mathcal{C}}) (X_{\mathcal{C}}^{-1})^T. \end{aligned} \quad (23)$$

We compare additive-noise and attenuator channel representations. In both cases, the dual map and the outcome rescaling map the covariance matrix Σ_{id} of the reference Gaussian POVM to the covariance matrix Σ_m of the target POVM:

$$\mathcal{C}^* : \quad \Sigma_{\text{id}}^{(\gamma)} \mapsto \Sigma_m^{(\gamma)} \quad (24)$$

A. Additive-noise channel

First, we consider the additive-noise (classical mixing) channel Φ_{γ} , characterized by the matrices [28]

$$X_{\Phi}^{(\gamma)} = \mathbb{1}, \quad Y_{\Phi}^{(\gamma)} \succeq 0, \quad \gamma \in \{\text{H}, \text{DH}\}. \quad (25)$$

The additive-noise channel is self-dual, $\Phi_{\gamma}^* = \Phi_{\gamma}$. It leaves the first moment unchanged and transforms the covariance matrix as

$$\Phi_{\gamma} : \Sigma \mapsto \Sigma + Y_{\Phi}^{(\gamma)}. \quad (26)$$

Equation (24) requires

$$\Phi_{\gamma} : \quad \Sigma_{\text{id}}^{(\gamma)} \mapsto \Sigma_m^{(\gamma)} = \Sigma_{\text{id}}^{(\gamma)} + Y_{\Phi}^{(\gamma)}, \quad (27)$$

and therefore

$$Y_{\Phi}^{(\gamma)} = \Sigma_m^{(\gamma)} - \Sigma_{\text{id}}^{(\gamma)} \equiv \Sigma_N^{(\gamma)}, \quad (28)$$

where $\Sigma_N^{(\gamma)}$ is the receiver excess-noise covariance matrix.

For homodyne detection, from Eq. (7) and (6), we have

$$\Sigma_N^{(\text{H})} = \Sigma_m^{(\text{H})} - \Sigma_{\text{id}}^{(\text{H})} = R(\phi) \text{diag}(\sigma_N, 0) R^T(\phi) \succeq 0, \quad (29)$$

where σ_N is given by Eq. (8).

For double-homodyne detection, Eqs. (15) and (18) give

$$\Sigma_N^{(\text{DH})}(r) = \Sigma_m^{(\text{DH})} - \Sigma_{\text{id}}^{(\text{DH})} = R(\phi) \begin{pmatrix} \sigma_N^{(1)}(r) & 0 \\ 0 & \sigma_N^{(2)}(r) \end{pmatrix} R^T(\phi) \succeq 0, \quad (30)$$

$$\sigma_N^{(1)}(r) = \delta_1 - e^{2r}, \quad \sigma_N^{(2)}(r) = \delta_2 - e^{-2r}. \quad (31)$$

The squeezing parameter r determines the entries of $\Sigma_N^{(\text{DH})}$. For every $r \in [r_2, r_1]$, the excess-noise covariance matrix is positive semidefinite, $\Sigma_N^{(\text{DH})}(r) \succeq 0$.

B. Attenuator channel

We shall move on to the attenuator channel $\mathcal{E}_\gamma$, characterized by matrices [28]

$$X_{\mathcal{E}}^{(\gamma)} = \cos \theta_\gamma \mathbb{1}, \quad Y_{\mathcal{E}}^{(\gamma)} = \sin^2 \theta_\gamma \Sigma_{\text{env}}^{(\gamma)}, \quad \gamma \in \{\text{H}, \text{DH}\}, \quad (32)$$

where $\Sigma_{\text{env}}^{(\gamma)}$ is the covariance matrix of the environmental input state.

Using Eqs. (22) and (24), we choose $\mathcal{E}_\gamma^*$ so that it maps the reference Gaussian POVM to the target POVM. In the rescaled outcome coordinates, its action on the covariance matrix is

$$\mathcal{E}_\gamma^* : \quad \Sigma_{\text{id}}^{(\gamma)} \mapsto \Sigma_m^{(\gamma)} = \frac{1}{\cos^2 \theta_\gamma} \Sigma_{\text{id}}^{(\gamma)} + \tan^2 \theta_\gamma \Sigma_{\text{env}}^{(\gamma)}, \quad \gamma \in \{\text{H}, \text{DH}\}. \quad (33)$$

For homodyne detection, we take the environmental input state to be the vacuum, with covariance matrix

$$\Sigma_{\text{env}}^{(\text{H})} = \mathbb{1}. \quad (34)$$

The attenuator transforms the measured quadrature according to

$$\hat{x}_{\phi, \text{out}} = \cos \theta_{\text{H}} \hat{x}_{\phi, \text{in}} + \sin \theta_{\text{H}} \hat{x}_{\phi, \text{env}}. \quad (35)$$

Let x_{raw} denote the outcome associated with $\hat{x}_{\phi, \text{out}}$. The rescaling $x = x_{\text{raw}} / \cos \theta_{\text{H}}$ corresponds to the rescaled observable

$$\frac{\hat{x}_{\phi, \text{out}}}{\cos \theta_{\text{H}}} = \hat{x}_{\phi, \text{in}} + \tan \theta_{\text{H}} \hat{x}_{\phi, \text{env}}. \quad (36)$$

Since the vacuum quadrature variance is one, the added variance is $\tan^2 \theta_{\text{H}}$. Exact equality with the target homodyne POVM therefore requires

$$\tan^2 \theta_{\text{H}} = \sigma_N. \quad (37)$$

Since $\sigma_x = 1 + \sigma_N$ by Eq. (8), Eq. (37) gives

$$\cos^2 \theta_{\text{H}} = \frac{1}{\sigma_x}, \quad \sin^2 \theta_{\text{H}} = \frac{\sigma_N}{\sigma_x}. \quad (38)$$

For double-homodyne detection, first suppose that $r_2 < r_1$, equivalently $\delta_1 \delta_2 > 1$, and take $r \in (r_2, r_1)$. In the attenuator representation studied here, we take the environmental input state to be a centered pure Gaussian state with covariance matrix

$$\Sigma_{\text{env}}^{(\text{DH})}(r) = R(\phi) \text{diag}(a(r), a(r)^{-1}) R^T(\phi), \quad a(r) > 0. \quad (39)$$

Equations (33), (18), and (15) give

$$\Sigma_m^{(\text{DH})} = R(\phi) \text{diag}(\delta_1, \delta_2) R^T(\phi) = R(\phi) \left\{ \frac{1}{\cos^2 \theta_{\text{DH}}(r)} \text{diag}(e^{2r}, e^{-2r}) + \tan^2 \theta_{\text{DH}}(r) \text{diag}(a(r), a(r)^{-1}) \right\} R^T(\phi), \quad (40)$$

$$\begin{cases} \delta_1 = \frac{e^{2r}}{\cos^2 \theta_{\text{DH}}(r)} + \tan^2 \theta_{\text{DH}}(r) a(r), \\ \delta_2 = \frac{e^{-2r}}{\cos^2 \theta_{\text{DH}}(r)} + \tan^2 \theta_{\text{DH}}(r) a(r)^{-1} \end{cases}. \quad (41)$$

Solving the system yields

$$\cos^2 \theta_{\text{DH}}(r) = \frac{e^{2r} + a(r)}{\delta_1 + a(r)} = \frac{\delta_1 e^{-2r} + \delta_2 e^{2r} - 2}{\delta_1 \delta_2 - 1}, \quad (42)$$

$$a(r) = \frac{\delta_1 - e^{2r}}{\delta_2 e^{2r} - 1}. \quad (43)$$

As $r \rightarrow r_2^+$ or $r \rightarrow r_1^-$, respectively, $a(r)$ or $a(r)^{-1}$ diverges while $\tan^2 \theta_{\text{DH}}(r) \rightarrow 0$. The products $\tan^2 \theta_{\text{DH}}(r)a(r)$ and $\tan^2 \theta_{\text{DH}}(r)a(r)^{-1}$ approach the required finite limits at r_2 and r_1 , respectively. For either endpoint $r_j \in \{r_2, r_1\}$, with the limit taken from the interval interior,

$$\lim_{r \rightarrow r_j} X_{\mathcal{E}}^{(\text{DH})}(r) = \mathbb{I}, \quad \lim_{r \rightarrow r_j} Y_{\mathcal{E}}^{(\text{DH})}(r) = \Sigma_N^{(\text{DH})}(r_j). \quad (44)$$

Thus, the attenuator channel converges to the corresponding additive-noise channel. If $r_1 = r_2$, then $\delta_1 \delta_2 = 1$, the target POVM is rank one, and both channel representations reduce to the identity channel.

IV. GAUSSIAN-MODULATED COHERENT-STATE PROTOCOL

In the Gaussian-modulated coherent-state (GMCS) protocol, Alice samples the complex amplitude $\alpha = \alpha_1 + i\alpha_2$ from the Gaussian prior:

$$p_A(\alpha) = \frac{1}{\pi V_A} \exp \left[-\frac{|\alpha|^2}{2V_A} \right]. \quad (45)$$

She then prepares the coherent state $|\alpha\rangle$. The states pass the composite Gaussian channel $\mathcal{N}_{A \rightarrow B} = \mathcal{L}_N \circ \mathcal{L}_{T_{\text{ch}}}$, where index indicates pure loss channel transmission coefficient, T_{ch} is the channel transmittance, $N = 1 + \xi_{\text{ch}}/(1 - T_{\text{ch}})$, and ξ_{ch} is the channel excess noise. Then, the receiver uses either homodyne or double-homodyne detection to measure the states. Mutual information between legitimate parties in each case is then given by the following expressions (see Ref. [25] for derivation)

$$I_{\text{AB}}^{(\text{H})} = \frac{1}{2} \log \left[1 + \frac{4T_{\text{ch}}V_A}{\sigma_x + \xi_{\text{ch}}} \right], \quad (46)$$

$$I_{\text{AB}}^{(\text{DH})} = \frac{1}{2} \sum_i \log \left[1 + \frac{4T_{\text{ch}}V_A}{2\sigma_i + \xi_{\text{ch}}} \right], \quad (47)$$

where σ_x and σ_i given by Eqs. (5) and (14), respectively, and upper indices hereinafter stand for homodyne and double-homodyne detection.

The covariance matrix of TMSV state with the Bob's mode transmitted through the noisy channel reads (see Ref. [30])

$$\mathcal{I}_A \otimes \mathcal{L}_N \circ \mathcal{L}_{T_{\text{ch}}} : \quad \Sigma_{\text{TMSV}} = \begin{pmatrix} V\mathbb{I} & \sqrt{V^2 - 1}\sigma_z \\ \sqrt{V^2 - 1}\sigma_z & V\mathbb{I} \end{pmatrix} \mapsto \Sigma_{\text{AB}} = \begin{pmatrix} V\mathbb{I} & c\sigma_z \\ c\sigma_z & V_B\mathbb{I} \end{pmatrix}, \quad (48)$$

where $\mathcal{I}_A$ is the identity channel acting on the subsystem A and the parameters of the covariance matrix Σ_{AB} are given by

$$V = 1 + 4V_A, \quad c = \sqrt{T_{\text{ch}}(V^2 - 1)}, \quad (49)$$

$$V_B = T_{\text{ch}}(V - 1) + 1 + \xi_{\text{ch}}. \quad (50)$$

We consider local action of either attenuator channel $\mathcal{E}_\gamma$ (32) or additive-noise channel Φ_γ (25), on second mode of TMSVS shared by Alice and Bob (48). In both cases, mutual information is given by $I_{\text{AB}}^{(\gamma)}$ (46), (47), as it is classical.

It follows that

$$\mathcal{I}_A \otimes \mathcal{E}_\gamma : \Sigma_{\text{AB}} \mapsto \Sigma_{\text{AB}}^{(\mathcal{E}_\gamma)} = \begin{pmatrix} V\mathbb{I} & c \cos \theta_\gamma \sigma_z \\ c \cos \theta_\gamma \sigma_z & \cos^2 \theta_\gamma \left(V_B + \tan^2 \theta_\gamma \Sigma_{\text{env}}^{(\gamma)} \right) \end{pmatrix}, \quad (51)$$

$$\mathcal{I}_A \otimes \Phi_\gamma : \Sigma_{\text{AB}} \mapsto \Sigma_{\text{AB}}^{(\Phi_\gamma)} = \begin{pmatrix} V\mathbb{I} & c\sigma_z \\ c\sigma_z & V_B\mathbb{I} + \Sigma_N^{(\gamma)} \end{pmatrix}, \quad \gamma \in \{\text{H}, \text{DH}\}. \quad (52)$$

When the channel $\mathcal{C}_\gamma \in \{\Phi_\gamma, \mathcal{E}_\gamma\}$ is under control of the eavesdropper, Eve holds a purification of the state $\hat{\rho}_{\text{AB}}^{(\mathcal{C}_\gamma)} = (\mathcal{I}_A \otimes \mathcal{C}_\gamma)(\hat{\rho}_{\text{AB}})$, and the receiver then applies the corresponding reference Gaussian POVM. In this scenario, the difference between the joint and conditional entropies

$$\chi_{\text{EB}}^{(\mathcal{C}_\gamma)} = S_{\text{E}} - S_{\text{E}|\text{B}} = S_{\text{AB}}^{(\mathcal{C}_\gamma)} - S_{\text{A}|\text{B}}^{(\mathcal{C}_\gamma)}, \quad \mathcal{C}_\gamma \in \{\Phi_\gamma, \mathcal{E}_\gamma\} \quad (53)$$

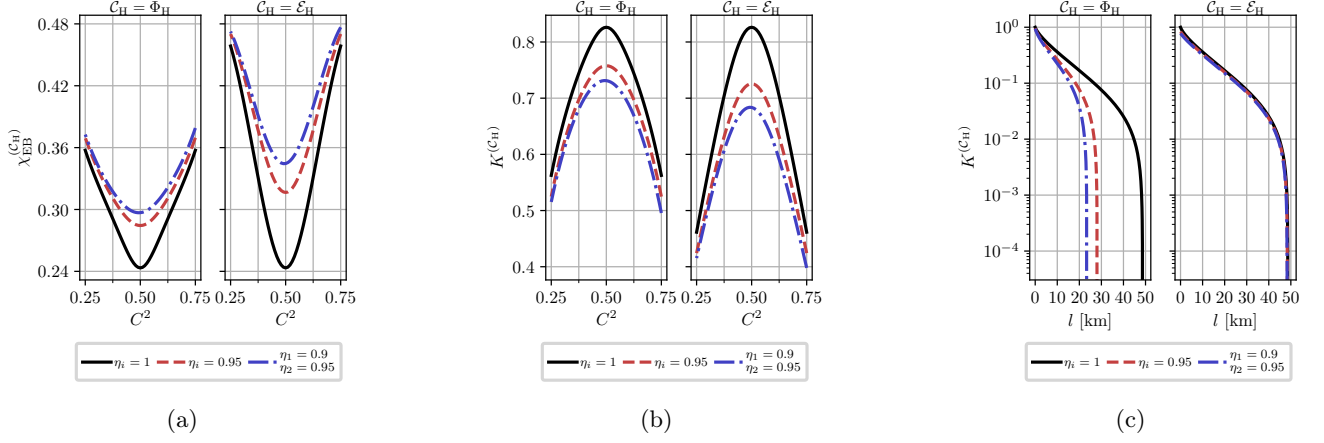


Figure 2: (a) Holevo information, $\chi_{EB}^{(C_H)}$ and (b) asymptotic secret fraction, $K^{(C_H)}$, as a function of the beam splitter transmission, C^2 , of the homodyne receiver computed at different values of the detector efficiencies for the additive-noise and attenuator channel representations, as indicated in the panel title. The parameters are: $V = 5$; $T_{ch} = 0.95$; $\xi_{ch} = 10^{-2}$; and $\beta = 0.95$. (c) $K^{(C_H)}$ versus transmission length l for the additive-noise and attenuator channel representations. The parameters are: $V = 5$; $\beta = 0.95$; $\xi_{ch} = 10^{-2}$; losses are 20 dB per 100 km.

gives the Holevo information. The joint entropy

$$S_{AB}^{(C_\gamma)} = g(\nu_1^{(C_\gamma)}) + g(\nu_2^{(C_\gamma)}), \quad (54)$$

$$g(\nu) = \frac{\nu + 1}{2} \log \frac{\nu + 1}{2} - \frac{\nu - 1}{2} \log \frac{\nu - 1}{2}$$

then can be expressed in terms of the symplectic eigenvalues of the covariance matrix (52) given by

$$\nu_{1,2}^{(C_\gamma)} = \sqrt{\Delta_{C_\gamma}/2 \pm \sqrt{\Delta_{C_\gamma}^2/4 - D_{C_\gamma}}}, \quad D_{C_\gamma} = \det \Sigma_{AB}^{(C_\gamma)}, \quad (55)$$

$$\Delta_{C_\gamma} = \det([\Sigma_{AB}^{(C_\gamma)}]_{11}) + \det([\Sigma_{AB}^{(C_\gamma)}]_{22}) + 2 \det([\Sigma_{AB}^{(C_\gamma)}]_{12}), \quad (56)$$

where $[\Sigma_{AB}^{(C_\gamma)}]_{ii}$ ($[\Sigma_{AB}^{(C_\gamma)}]_{ij}$) are diagonal (off-diagonal) blocks of $\Sigma_{AB}^{(C_\gamma)}$.

Conditional entropies $S_{E|B} = S_{A|B}^{(\gamma)}$ are calculated from respective conditional covariance matrices,

$$S_{A|B}^{(\gamma)} = g(\nu_3^{(\gamma)}), \quad \nu_3^{(\gamma)} = \sqrt{\det \Sigma_{A|B}^{(\gamma)}}, \quad (57)$$

$$\Sigma_{A|B}^{(\gamma)} = V \mathbb{1} - c^2 \sigma_z (V_B \mathbb{1} + \Sigma_m^{(\gamma)})^{-1} \sigma_z. \quad (58)$$

Direct computations show that these conditional matrices, and therefore their symplectic eigenvalues, are independent of the channel representation. The symplectic eigenvalues are

$$\nu_3^{(H)} = \sqrt{V \left(V - \frac{c^2}{V_B + \sigma_N} \right)}, \quad (59)$$

$$\nu_3^{(DH)} = V \sqrt{\frac{(V_B + \delta_1 - \frac{c^2}{V})(V_B + \delta_2 - \frac{c^2}{V})}{(V_B + \delta_1)(V_B + \delta_2)}}, \quad (60)$$

From mutual information and Holevo information evaluation of the asymptotic secret fraction is done per Devetak-Winter formula [31]

$$K^{(C_\gamma)} = \beta I_{AB}^{(C_\gamma)} - \chi_{EB}^{(C_\gamma)}, \quad \gamma \in \{H, DH\}, C_\gamma \in \{\Phi_\gamma, \mathcal{E}_\gamma\}. \quad (61)$$

The additive-noise symplectic eigenvalues obtained from Eq. (55) do not simplify further. The following sections give analytic expressions only for the attenuator representation.

A. Homodyne detection

For attenuator channel, from Eq. (51) and (38) we have

$$\Sigma_{AB}^{(\mathcal{E}_H)} = \begin{pmatrix} V\mathbb{1} & \frac{c}{\sqrt{\sigma_x}}\sigma_z \\ \frac{c}{\sqrt{\sigma_x}}\sigma_z & \frac{1}{\sigma_x}(V_B + \sigma_N)\mathbb{1} \end{pmatrix} \equiv \begin{pmatrix} V\mathbb{1} & \tilde{c}\sigma_z \\ \tilde{c}\sigma_z & \tilde{V}_B\mathbb{1} \end{pmatrix}, \quad (62)$$

$$\tilde{V}_B \equiv T(V - 1) + 1 + \xi, \quad (63)$$

$$\tilde{c} \equiv \sqrt{T(V^2 - 1)}, \quad (64)$$

where we defined scaled Bob's variance and scaled correlations between Alice and Bob $\tilde{V}_B$ and $\tilde{c}$, using total channel transmission T that consists of channel transmission T_{ch} and detection losses $\frac{1}{\sigma_x}$, as well as scaled noise variance ξ ,

$$T = T_{\text{ch}} \cdot \frac{1}{\sigma_x}, \quad (65a)$$

$$\xi \equiv \xi_{\text{ch}} \cdot \frac{1}{\sigma_x}. \quad (65b)$$

It follows that for homodyne detection the effect of measurement asymmetry is equivalent to scaling both channel transmission (65a) and channel noise (65b) by a factor of σ_x^{-1} .

Symplectic eigenvalues of $\Sigma_{AB}^{(\mathcal{E}_H)}$, $\nu_{1,2}^{(\mathcal{E}_H)}$, can be computed using formulas [2]:

$$\nu_{1,2}^{(\mathcal{E}_H)} = \frac{z \pm [\tilde{V}_B - V]}{2}, \quad z = \sqrt{(V + \tilde{V}_B)^2 - 4\tilde{c}^2}, \quad (66)$$

and then substituted in Eq. (54) to calculate joint entropy $S_{AB}^{(\mathcal{E}_H)}$.

Fig. 2 illustrates the effects of homodyne measurement asymmetry on Holevo information (53) and asymptotic secret fraction (61) versus beam splitter transmission C^2 , as well as secret fraction versus transmission length, for both additive-noise (52) and attenuator (51) channels. While channel noise is reduced in this setup (65b), so is channel transmission (65a), and, since transmission is scaled by modulation variance $V \gg 1$ in every expression, as well as reduced mutual information with greater asymmetry per Eq. (46), attenuator channel has negative effect. As shown in Fig. 2a and 2b, the attenuator channel results in higher Holevo information, and thus lower secret fraction, at high transmission for fixed asymmetry parameters. However, Fig. 2c reveals that the attenuator channel achieves significantly greater reach (maximum transmission length at which a positive key rate is obtained) than the additive-noise channel.

B. Double-homodyne detection

For attenuator channel, from Eq. (51) and (42) we have

$$\Sigma_{AB}^{(\mathcal{E}_{\text{DH}})} = \begin{pmatrix} V\mathbb{1} & c \cos \theta_{\text{DH}}(r) \sigma_z & 0 \\ c \cos \theta_{\text{DH}}(r) \sigma_z & \cos^2 \theta_{\text{DH}}(r) \left(V_B + \delta_1 - \frac{e^{2r}}{\cos^2 \theta_{\text{DH}}(r)} \right) & 0 \\ 0 & 0 & \cos^2 \theta_{\text{DH}}(r) \left(V_B + \delta_2 - \frac{e^{-2r}}{\cos^2 \theta_{\text{DH}}(r)} \right) \end{pmatrix} \quad (67)$$

Symplectic eigenvalues of this matrix can be calculated using invariants similarly to Eq. (55). It is obvious that these symplectic eigenvalues, and thereby joint entropy, is explicitly depended on squeezing parameter r (19), like in the case of additive-noise channel (see Ref. [25]).

Fig. 3 illustrates dependence of Holevo information on the squeezing parameter, for both additive-noise and attenuator channels. Referring to Fig. 3, in the limiting case of ideal double-homodyne detection with $\delta_1 = \delta_2 = 1$, only one value of $r = 0$ is allowed (see Ref. [25]), with equal values of Holevo information for both channels (black dot on plot).

When $\delta_1 = \delta_2 > 1$, dependence of Holevo information on squeezing parameter is symmetrical around $r = 0$ for both channels. For additive-noise channel, $r = 0$ is the point of maximal Holevo information, while for attenuator channel $r = 0$ corresponds to minimal Holevo information. Moreover, while Holevo information is undefined at $r = r_{2,1}$ (see Eq. (42)), the values tend to the same values of joint entropy in the case of additive-noise channel, as signified by color-matched white-filled dots in Fig. 3.

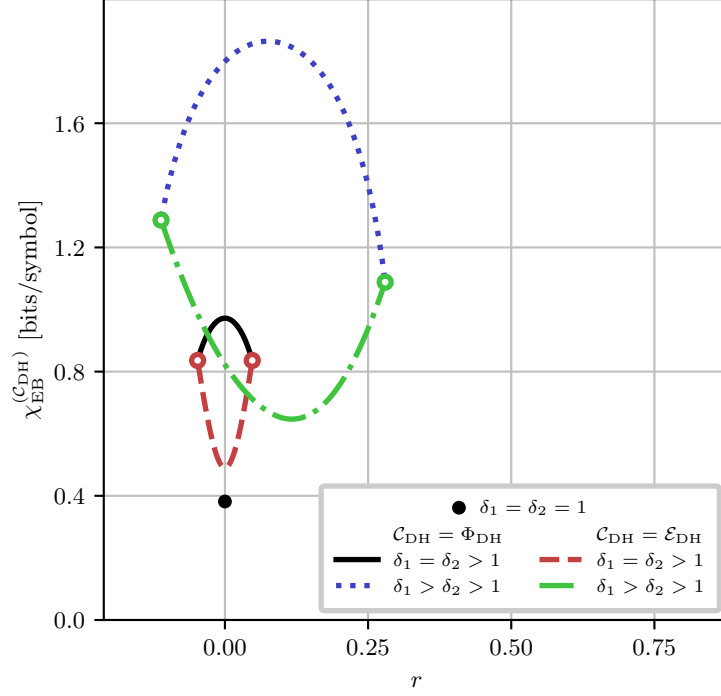


Figure 3: Holevo information, $\chi_{\text{EB}}^{(\text{DH}; C_{\text{DH}})}$, as a function of the squeezing parameter r at different values of $\delta_{1,2}$. Black dot corresponds to the case of perfect measurements. Solid and dash-dotted curves show the additive-noise representation; dashed and dotted curves show the attenuator representation. Color-matched white-filled dots mark points where Holevo information is undefined at the boundaries of the r interval (19) (see Eq. (43) and main text). Values of $\delta_{1,2}$ are: $\delta_{1,2} = 1.1$ for solid and dashed curves, $\delta_1 = 1.75, \delta_2 = 1.25$ for dashdotted and dotted curves. The channel parameters are: $V = 5$; $T_{\text{ch}} = 0.95$; and $\xi_{\text{ch}} = 10^{-2}$.

For the asymmetric parameter sets considered, where $\delta_1 \neq \delta_2$, the Holevo information for both channel representations becomes asymmetric about its extrema, resulting in unequal joint entropies at $r = r_{2,1}$. For these parameter sets, the Holevo information is concave in r for the additive-noise channel and convex in r for the attenuator channel.

Fig. 4 illustrates lower and upper bounds of the asymptotic secret fraction versus transmission length, for both additive-noise (52) and attenuator (51) channels. Note that for chosen $\sigma_x^{(i)}$ lower bounds for both channels coincide, because for $\sigma_x^{(i)} \approx 1$ maximum of joint entropy for both channel representations is either at $r = r_1$ or $r = r_2$. Moreover, the upper bound for additive-noise channel in Fig. 4a is very close (0.2 km difference) to the lower bound.

As seen from Fig. 4, difference between lower and upper bound of the asymptotic secret fraction is greater in

Fig. 5 illustrates the dependence of Holevo information (53) and asymptotic secret fraction (61) on the signal beam splitter transmission C_S^2 , as well as secret fraction versus transmission length if the receiver uses double-homodyne detection, for both additive-noise (52) and attenuator (51) channels. Unlike the homodyne detection case (see Fig. 2), Holevo information is significantly lower for attenuator channel, except for the ideal measurements case ($\eta_{1,2}^{(1,2)} = 1$ and whole C_S^2 axis, unlike in homodyne case where ideal measurements are strictly $\eta_{1,2} = 1$ and $C^2 = 0.5$) as both channels have no effect and thereby have equal joint entropies (black curve). High difference in Holevo information is due to the dependence of joint entropy on r (see Fig. 3) and fixing $r = \frac{r_2 + r_1}{2}$ for both curves, effectively minimizing Holevo information for attenuator channel and maximizing it for additive-noise channel.

Analogously to the case of homodyne detection (see Fig. 2c), in the case of double-homodyne detection the attenuator channel has greater reach than the additive-noise channel, as illustrated by Fig. 5c.

V. DISCUSSION AND CONCLUSION

In this paper, we have constructed and compared additive-noise and attenuator channel representations of the target homodyne and double-homodyne POVMs. Our approach requires the dual map of the selected Gaussian channel,

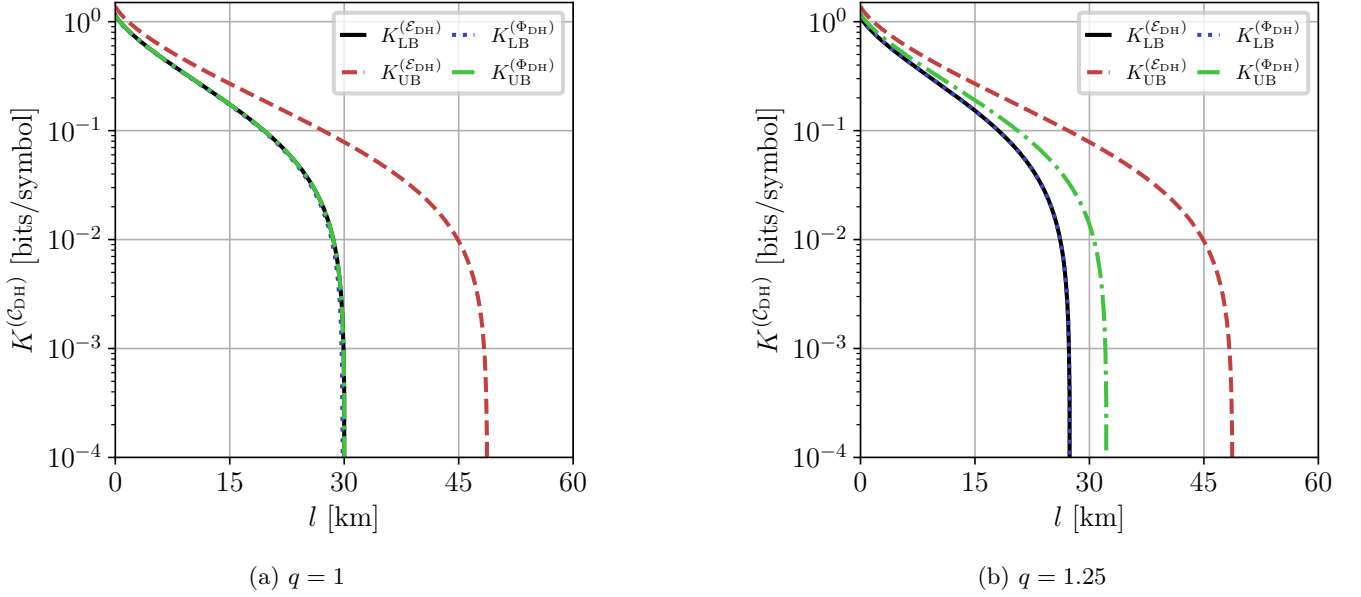


Figure 4: Asymptotic secret fraction $K^{(C_{DH})}$ versus channel length l , computed for fixed $\sigma_x^{(i)} = 1.05$ (see Eq. (14)) and various imbalance ratios $q \equiv \frac{S_S^2}{C_S^2}$ for the additive-noise and attenuator channel representations (see Fig. 3). The parameters are: $V = 5$; $\beta = 0.95$; $\xi_{ch} = 10^{-2}$; the fiber loss is 20 dB per 100 km.

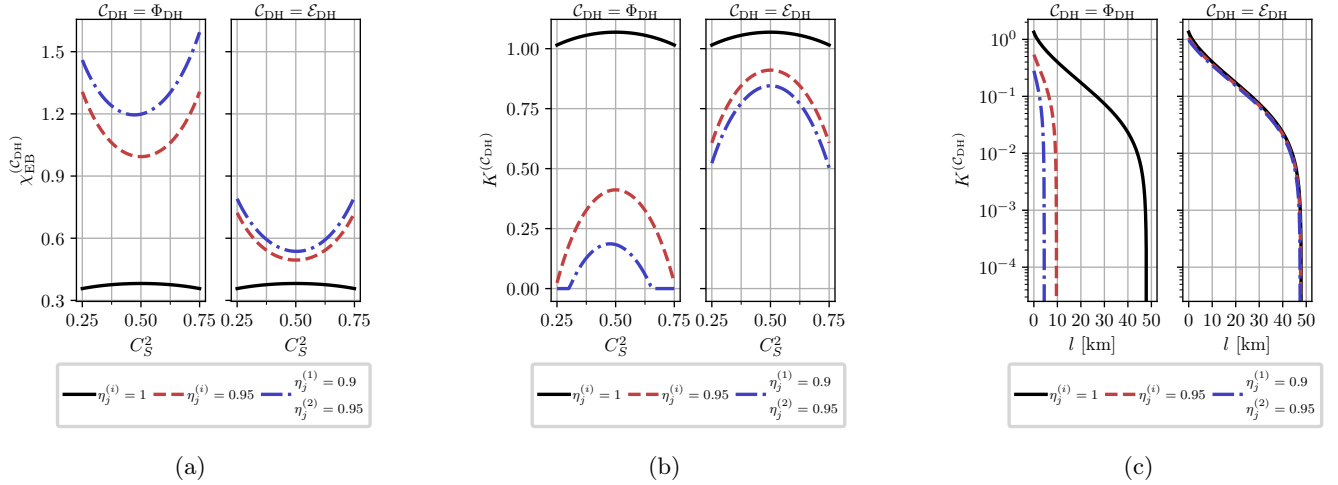


Figure 5: (a) Holevo information, $\chi_{EB}^{(C_{DH})}$ and (b) asymptotic secret fraction, $K^{(C_{DH})}$, as a function of the signal beam splitter transmission, C_S^2 , of the double-homodyne receiver computed at different values of the detector efficiencies for the additive-noise and attenuator channel representations, as indicated in the panel title. The parameters are: $V = 5$; $T_{ch} = 0.95$; $\xi_{ch} = 10^{-2}$; and $\beta = 0.95$. (c) $K^{(C_{DH})}$ versus transmission length l for the additive-noise and attenuator channel representations. The parameters are: $V = 5$; $\beta = 0.95$; $\xi_{ch} = 10^{-2}$; losses are 20 dB per 100 km.. For both channel representations $r = \frac{r_2 + r_1}{2}$ (see main text).

together with a deterministic invertible rescaling of the measurement outcomes, to reproduce the target POVM. At the level of covariance matrices, this condition is expressed by Eq. (24). It constrains the channel parameters, which are then determined by the receiver parameters and, for double-homodyne detection, by the squeezing parameter r .

After the deterministic invertible rescaling of the measurement outcomes, both channels reproduce the same conditional statistics of Bob's outcomes. They therefore give the same Alice-Bob mutual information. For fixed source and channel parameters, the conditional covariance matrix in Eq. (58), and hence the conditional entropy $S_{A|B}^{(\gamma)}$, are the same for both channels because the detector imperfections enter the conditional state through the same target

POVM. The difference between the security estimates originates from the joint state of Alice and Bob. The additive-noise and attenuator channels generally produce different joint covariance matrices and, consequently, different joint entropies. In the untrusted-detector model considered here, Eve holds a purification of the Alice–Bob state after the effective receiver channel. The corresponding Holevo information (53) can therefore depend on the selected channel while Bob’s observed statistics remain unchanged.

For homodyne detection, we take the environmental mode of the attenuator channel to be in the vacuum state, so that the detector imperfections are represented by a pure-loss channel. This channel rescales both the effective transmission and the output-referred excess noise by the same factor, as shown in Eq. (65).

The numerical results in Fig. 2 show that the additive-noise and attenuator channels give different security estimates. At high channel transmittance and for the parameters considered, the attenuator channel gives a larger Holevo information and a lower asymptotic secret fraction. At longer distances, however, the secret fraction remains positive up to a greater channel length than in the additive-noise channel. Therefore, neither channel can be described as strictly more or less pessimistic.

A distinguishing feature of double-homodyne detection is the nonuniqueness of the channel representation of its target POVM. The covariance matrix of the physical double-homodyne measurement is fixed by the receiver parameters and does not depend on the squeezing parameter r . The effects of the reference Gaussian POVM are proportional to projectors onto displaced squeezed states and depend on r . The target POVM is obtained by applying the dual map of the selected Gaussian channel to the reference Gaussian POVM, together with the outcome rescaling. The requirement that the excess-noise covariance matrix be positive semidefinite restricts r to the interval given by Eq. (19).

For the additive-noise channel, variation of r redistributes the measurement noise between the two conjugate quadratures according to Eq. (31). For the attenuator channel, the same variation changes both the attenuation coefficient and the covariance matrix of the environmental state, as described by Eqs. (42) and (43). In the attenuator channel considered here, the generally anisotropic measurement noise is reproduced by an environmental mode in an anisotropic Gaussian state, which was chosen to be pure.

When the admissible interval for r has nonzero width, one additive-noise quadrature variance vanishes at each boundary. The environmental state in the attenuator channel then approaches an infinitely squeezed limit, with one quadrature variance tending to zero and the conjugate variance diverging. The attenuator result approaches the corresponding additive-noise result. If the interval collapses to a single value, the target POVM is rank one and both channel representations reduce to the identity channel.

Figure 3 shows that the Holevo information depends on r in qualitatively different ways for the two channel representations. For the parameter sets considered, this dependence is concave for the additive-noise channel and convex for the attenuator channel. When the measurement noise is symmetric between the two quadratures, both curves are symmetric about $r = 0$. In this case, the coherent state, given by $r = 0$, maximizes the Holevo information for the additive-noise channel and minimizes it for the attenuator channel. For the asymmetric case considered, this symmetry is lost and the extrema are shifted away from $r = 0$.

The lower and upper secret-fraction curves in Fig. 4 describe the range obtained by varying r over its admissible interval for each channel representation. For the near-ideal quadrature variances used in Fig. 4, the lower bounds of both channels nearly coincide because the Holevo-information maximum occurs at, or close to, the interval boundaries. The upper bounds become more clearly separated when the imbalance of the signal beam splitter is increased.

The comparison shown in Fig. 5 was performed at the midpoint of the admissible interval of r . In the symmetric case, this value coincides with the maximum of the additive-noise Holevo information and the minimum of the attenuator Holevo information. The pronounced separation between the secret fractions in Fig. 5 therefore reflects both the selected channel representation and the selected squeezing parameter.

These results clarify the role of the chosen channel representation when receiver noise is treated as untrusted. The additive-noise and attenuator channels reproduce the same receiver statistics after outcome rescaling. They generally yield different joint states of Alice and Bob. In the untrusted-detector model, Eve holds a purification of the state produced by the selected channel. Choosing a channel representation therefore introduces an assumption about the effective quantum map assigned to receiver asymmetry. A security estimate should not be based on a particular channel representation solely because it gives a larger secret fraction. The selected representation should either be justified from the physical structure of the receiver or the dependence of the estimate on the channels included in the analysis should be reported explicitly. The present comparison is restricted to the additive-noise channel and to attenuator channels whose environmental modes are in pure Gaussian states. Mixed environmental states and other Gaussian channel representations compatible with the same receiver statistics were not considered.

Appendix A extends the receiver model to include independent pure losses at the output ports of a lossless beam splitter. These losses can be incorporated into the effective photodetector efficiencies. By contrast, a general lossy asymmetric beam splitter with unequal complex transmission and reflection amplitudes does not, in general, preserve the simple quadrature form of the receiver response and cannot be included by merely replacing the detector efficiencies.

The security analysis was performed in the asymptotic limit for the GMCS CV-QKD protocol under collective

Gaussian attacks, with the modeled receiver imperfections treated within an untrusted-detector model. Trusted receiver noise, finite-size corrections, active detector attacks, and uncertainties in the calibration parameters require separate treatment. Within these assumptions, equivalent receiver statistics can lead to different estimates of Eve's information when different channel representations are used within the untrusted-detector model.

ACKNOWLEDGEMENTS

The work was supported by the Russian Science Foundation (project No. [24-11-00398](#)).

Appendix A: Lossy beam splitter

In this appendix, we show that applying independent pure loss channels at the output ports of a lossless beam splitter yields the exact same transformation as a single formally defined lossy beam splitter.

First, consider the case of homodyne detection. The lossless unbalanced beam splitter mixes the input signal mode $\hat{a}$ and the local oscillator mode $\hat{a}_L$. This transformation is described by the unitary rotation matrix $\mathbf{U}_{\text{BS}}$,

$$\begin{pmatrix} \hat{a}_1 \\ \hat{a}_2 \end{pmatrix} = \mathbf{U}_{\text{BS}} \begin{pmatrix} \hat{a} \\ \hat{a}_L \end{pmatrix} = \begin{pmatrix} C & S \\ -S & C \end{pmatrix} \begin{pmatrix} \hat{a} \\ \hat{a}_L \end{pmatrix} \\ = \begin{pmatrix} C\hat{a} + S\hat{a}_L \\ -S\hat{a} + C\hat{a}_L \end{pmatrix}, \quad (\text{A1})$$

where C and S are the transmission and reflection coefficients of the beam splitter BS, respectively.

We model the subsequent losses by passing each intermediate output mode $\hat{a}_{1,2}$ through a virtual beam splitter of transmittance $\tau_{1,2}$. This mixes each signal with an independent environmental vacuum mode $\hat{v}_{1,2}$, where $[\hat{v}_i, \hat{v}_j^\dagger] = \delta_{ij}$,

$$\begin{pmatrix} \hat{b}_1 \\ \hat{c}_1 \end{pmatrix} = \begin{pmatrix} \sqrt{\tau_1} & -\sqrt{1-\tau_1} \\ \sqrt{1-\tau_1} & \sqrt{\tau_1} \end{pmatrix} \begin{pmatrix} \hat{a}_1 \\ \hat{v}_1 \end{pmatrix}, \quad (\text{A2})$$

$$\begin{pmatrix} \hat{b}_2 \\ \hat{c}_2 \end{pmatrix} = \begin{pmatrix} \sqrt{\tau_2} & -\sqrt{1-\tau_2} \\ \sqrt{1-\tau_2} & \sqrt{\tau_2} \end{pmatrix} \begin{pmatrix} \hat{a}_2 \\ \hat{v}_2 \end{pmatrix}. \quad (\text{A3})$$

Tracing out the lost environmental modes $\hat{c}_{1,2}$, the remaining physical output modes $\hat{b}_{1,2}$ expand to

$$\begin{pmatrix} \hat{b}_1 \\ \hat{b}_2 \end{pmatrix} = \begin{pmatrix} \sqrt{\tau_1}C & \sqrt{\tau_1}S \\ -\sqrt{\tau_2}S & \sqrt{\tau_2}C \end{pmatrix} \begin{pmatrix} \hat{a} \\ \hat{a}_L \end{pmatrix} + \begin{pmatrix} -\sqrt{1-\tau_1}\hat{v}_1 \\ -\sqrt{1-\tau_2}\hat{v}_2 \end{pmatrix}. \quad (\text{A4})$$

From Eq. (A4), we extract the effective scattering matrix $\mathbf{T}_{\text{BS}}$ and the fluctuation noise operators $\hat{F}_{1,2}$ representing the device excitations

$$\mathbf{T}_{\text{BS}} = \begin{pmatrix} \sqrt{\tau_1}C & \sqrt{\tau_1}S \\ -\sqrt{\tau_2}S & \sqrt{\tau_2}C \end{pmatrix}. \quad (\text{A5})$$

It is not difficult to show that to preserve the Canonical Commutation Relations for the output fields, $[\hat{b}_i, \hat{b}_j^\dagger] = \delta_{ij}$, the noise operators $\hat{F}_{1,2}$ must satisfy

$$[\hat{F}_i, \hat{F}_j^\dagger] = \delta_{ij} - \sum_k T_{ik} T_{jk}^*, \quad (\text{A6})$$

where T_{ik} are the elements of the scattering matrix $\mathbf{T}_{\text{BS}}$ (A5). Evaluating this constraint explicitly using the elements of $\mathbf{T}_{\text{BS}}$ yields:

$$[\hat{F}_1, \hat{F}_1^\dagger] = 1 - \tau_1, \quad (\text{A7})$$

$$[\hat{F}_2, \hat{F}_2^\dagger] = 1 - \tau_2, \quad (\text{A8})$$

$$[\hat{F}_1, \hat{F}_2^\dagger] = 0, \quad (\text{A9})$$

which matches the commutators calculated from Eq. (A4).

We now connect this setup to the formal, standard mathematical treatment of lossy beam splitters [32, 33]. This framework models loss via an interaction matrix $\mathbf{A}$ acting on ancillary modes, which must obey the positivity condition $\mathbf{1} - \mathbf{A}\mathbf{A}^\dagger \geq 0$ to satisfy unitarity. Note that the extracted scattering matrix $\mathbf{T}_{\text{BS}}$ (A5) can be factored into a product of a diagonal scaling matrix and the lossless unitary,

$$\begin{aligned}\mathbf{T}_{\text{BS}} &= \begin{pmatrix} \sqrt{\tau_1} & 0 \\ 0 & \sqrt{\tau_2} \end{pmatrix} \begin{pmatrix} C & S \\ -S & C \end{pmatrix} \\ &= \begin{pmatrix} \sqrt{\tau_1} & 0 \\ 0 & \sqrt{\tau_2} \end{pmatrix} \mathbf{U}_{\text{BS}}.\end{aligned}\quad (\text{A10})$$

By defining the positive semi-definite matrix representing the loss properties as $\mathbf{A}\mathbf{A}^\dagger = \text{diag}(1 - \tau_1, 1 - \tau_2)$, it follows that:

$$\sqrt{\mathbf{1} - \mathbf{A}\mathbf{A}^\dagger} = \begin{pmatrix} \sqrt{\tau_1} & 0 \\ 0 & \sqrt{\tau_2} \end{pmatrix}. \quad (\text{A11})$$

Substituting this back into Eq. (A10) yields the polar decomposition of the scattering matrix of a lossy beam splitter in its generalized form [33]:

$$\mathbf{T}_{\text{BS}} = \sqrt{\mathbf{1} - \mathbf{A}\mathbf{A}^\dagger} \mathbf{U}_{\text{BS}}. \quad (\text{A12})$$

Because the matrix $\mathbf{A}\mathbf{A}^\dagger$ representing the losses is diagonal, it is straightforward to see that to account for lossy beam splitter in a homodyne setup it is sufficient to simply rescale the quantum efficiencies of detection: $\eta_{1,2} \rightarrow \tau_{1,2}\eta_{1,2}$. The same argument would apply in the case of double-homodyne detection.

We conclude this appendix by discussing the generalized (asymmetrical) beam splitter model where input ports of the beam splitter are characterized by different set of transmission and reflection coefficients. In this case, the non-unitary scattering matrix is written in the form [34]

$$\mathbf{S} = \begin{pmatrix} t & \rho \\ r & \tau \end{pmatrix}, \quad \mathbf{S}\mathbf{S}^\dagger \leq \mathbf{1}, \quad (\text{A13})$$

where t, τ (r, ρ) are complex transmission (reflection) coefficients. While it can be factored into lossy and unitary parts similarly to (A12), **this case cannot be covered by the formulas after some replacement, because it ruins quadrature representation:**

$$|\alpha, \alpha_L\rangle \mapsto |\alpha_1, \alpha_2\rangle, \quad (\text{A14})$$

$$\alpha_1 = t\alpha + \rho\alpha_L, \quad \alpha_2 = r\alpha + \tau\alpha_L, \quad (\text{A15})$$

$$\mathbf{P}_G(\mu) = G(\mu - \mu_G; \sigma_G), \quad (\text{A16})$$

$$\begin{aligned}\sigma_G &= \eta_1 |\alpha_1|^2 + \eta_2 |\alpha_2|^2 \\ &= (\eta_1 |\rho|^2 + \eta_2 |\tau|^2) |\alpha_L|^2 + (\eta_1 |t|^2 + \eta_2 |r|^2) |\alpha|^2 \\ &\quad + 2 \text{Re}[(\eta_1 t \rho^* + \eta_2 r \tau^*) \alpha \alpha_L^*] \\ &\approx (\eta_1 |\rho|^2 + \eta_2 |\tau|^2) |\alpha_L|^2 + 2 \text{Re}[(\eta_1 t \rho^* + \eta_2 r \tau^*) \alpha \alpha_L^*],\end{aligned}\quad (\text{A17})$$

$$\begin{aligned}\mu_G &= \eta_1 |\alpha_1|^2 - \eta_2 |\alpha_2|^2 \\ &= (\eta_1 |\rho|^2 - \eta_2 |\tau|^2) |\alpha_L|^2 + (\eta_1 |t|^2 - \eta_2 |r|^2) |\alpha|^2 \\ &\quad + 2 \text{Re}[(\eta_1 t \rho^* - \eta_2 r \tau^*) \alpha \alpha_L^*] \\ &\approx (\eta_1 |\rho|^2 - \eta_2 |\tau|^2) |\alpha_L|^2 + 2 \text{Re}[(\eta_1 t \rho^* - \eta_2 r \tau^*) \alpha \alpha_L^*].\end{aligned}\quad (\text{A18})$$

Appendix B: Fluctuating Local Oscillator

In this Appendix, we extend the theoretical treatment of homodyne measurements to account for classical intensity fluctuations of the local oscillator (LO).

We start from the Gaussian approximation of the photon count difference probability distribution in homodyne scheme [25]

$$P_G(\mu) = G(\mu - \mu_G; \sigma_G), \quad (B1)$$

$$\sigma_G = \eta_1 |\alpha_1|^2 + \eta_2 |\alpha_2|^2 \approx (\eta_1 S^2 + \eta_2 C^2) |\alpha_L|^2, \quad (B2)$$

$$\mu_G = \eta_1 |\alpha_1|^2 - \eta_2 |\alpha_2|^2 \approx (\eta_1 S^2 - \eta_2 C^2) |\alpha_L|^2 + CS(\eta_1 + \eta_2) |\alpha_L| \langle \hat{x}_\phi \rangle, \quad (B3)$$

where

$$G(x; \sigma) \equiv \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{x^2}{2\sigma}\right). \quad (B4)$$

We assume that the LO amplitude fluctuates around its mean value α_L such that

$$\alpha_L = \alpha_L + \delta\alpha_L, \quad \delta\alpha_L \sim G(\delta\alpha_L, \sigma_L) \quad (B5)$$

where σ_{LO} represents the variance of the LO amplitude fluctuations.

To find the total probability distribution $P_{\text{tot}}(\mu)$ that accounts for the local oscillator fluctuations, we perform an ensemble average by integrating the conditional distribution $P_G(\mu)$ over the Gaussian prior of the LO amplitude.

$$P_{\text{total}}(\mu) = \int d\delta\alpha_L P_G(\mu|\delta\alpha_L) P_G(\delta\alpha_L) = \quad (B6)$$

S

-
- [1] S. L. Braunstein and P. van Loock, Quantum information with continuous variables, *Rev. Mod. Phys.* **77**, 513 (2005).
 - [2] C. Weedbrook, S. Pirandola, R. García-Patrón, N. J. Cerf, T. C. Ralph, J. H. Shapiro, and S. Lloyd, Gaussian quantum information, *Rev. Mod. Phys.* **84**, 621 (2012).
 - [3] A. S. Holevo, Structure of a general quantum gaussian observable, *Proc. Steklov Inst. Math.* **313**, 70 (2021).
 - [4] G. M. D'Ariano, M. G. Paris, and M. F. Sacchi, Quantum tomography, *Advances in imaging and electron physics* **128**, S1076 (2003).
 - [5] A. I. Lvovsky and M. G. Raymer, Continuous-variable optical quantum-state tomography, *Rev. Mod. Phys.* **81**, 299 (2009).
 - [6] Y.-L. Zhang, C.-S. Yang, C.-L. Zou, T. Xia, G.-C. Guo, and X.-B. Zou, Quantum state preparation of an atomic ensemble via cavity-assisted homodyne measurement, *Journal of Physics B: Atomic, Molecular and Optical Physics* **52**, 215003 (2019).
 - [7] F. Grosshans and P. Grangier, Continuous variable quantum cryptography using coherent states, *Phys. Rev. Lett.* **88**, 057902 (2002).
 - [8] S. Pirandola, U. L. Andersen, L. Banchi, M. Berta, D. Bunandar, R. Colbeck, D. Englund, T. Gehring, C. Lupo, C. Ottaviani, J. L. Pereira, M. Razavi, J. S. Shaari, M. Tomamichel, V. C. Usenko, G. Vallone, P. Villoresi, and P. Wallden, Advances in quantum cryptography, *Adv. Opt. Photon.* **12**, 1012 (2020).
 - [9] Y. Zhang, Y. Bian, Z. Li, S. Yu, and H. Guo, Continuous-variable quantum key distribution system: Past, present, and future, *Applied Physics Reviews* **11**, 011318 (2024).
 - [10] E. Arthurs and J. L. Kelly, Jr., On the simultaneous measurement of a pair of conjugate observables, *Bell System Technical Journal* **44**, 725 (1965).
 - [11] A. S. Holevo, On the mathematical theory of quantum communication channels, *Problemy Peredachi Informatsii* **8**, 62 (1972), (in Russian).
 - [12] A. S. Holevo, Statistical Problems in Quantum Physics, in *Proceedings of the Second Japan-USSR Symposium on Probability Theory*, Lecture Notes in Mathematics, Vol. 330 (Springer, 1973) pp. 104–119.
 - [13] S. T. Ali and G. G. Emch, Fuzzy observables in quantum mechanics, *Journal of Mathematical Physics* **15**, 176 (1974).
 - [14] H. P. Yuen and V. W. S. Chan, Noise in homodyne and heterodyne detection, *Opt. Lett.* **8**, 177 (1983).
 - [15] P. Lahti, J.-P. Pellonpää, and J. Schultz, Realistic eight-port homodyne detection and covariant phase space observables, *Journal of Modern Optics* **57**, 1171 (2010).
 - [16] A. Barchielli and A. Santamato, Eight-port homodyne detector: The effect of imperfections on quantum random-number generation and on detection of quadratures, *Physical Review A* **106**, 10.1103/physreva.106.022620 (2022).
 - [17] D. Pereira, M. Almeida, M. Facão, A. N. Pinto, and N. A. Silva, Impact of receiver imbalances on the security of continuous variables quantum key distribution, *EPJ Quantum Technology* **8**, 10.1140/epjqt/s40507-021-00112-z (2021).
 - [18] C. Lupo and Y. Ouyang, Quantum key distribution with nonideal heterodyne detection: Composable security of discrete-modulation continuous-variable protocols, *PRX Quantum* **3**, 010341 (2022).
 - [19] J. Mi, Y. Bian, L. Fan, S. Yu, and Y. Zhang, Practical no-switching continuous-variable quantum key distribution with biased quadrature detection, *Phys. Rev. Applied* **21**, 064059 (2024).

- [20] F. Caruso, V. Giovannetti, and A. S. Holevo, One-mode bosonic gaussian channels: a full weak-degradability classification, *New Journal of Physics* **8**, 310–310 (2006).
- [21] J. Lodewyck, M. Bloch, R. García-Patrón, S. Fossier, E. Karpov, E. Diamanti, T. Debuisschert, N. J. Cerf, R. Tualle-Broui, S. W. McLaughlin, and P. Grangier, Quantum key distribution over 25 km with an all-fiber continuous-variable system, *Phys. Rev. A* **76**, 042305 (2007).
- [22] V. C. Usenko and R. Filip, Trusted noise in continuous-variable quantum key distribution: A threat and a defense, *Entropy* **18**, 10.3390/e18010020 (2016).
- [23] F. Laudenbach and C. Pacher, Analysis of the trusted-device scenario in continuous-variable quantum key distribution, *Advanced Quantum Technologies* **2**, 1900055 (2019).
- [24] R. Namiki, A. Kitagawa, and T. Hirano, Secret key rate of a continuous-variable quantum-key-distribution scheme when the detection process is inaccessible to eavesdroppers, *Phys. Rev. A* **98**, 042319 (2018).
- [25] A. S. Naumchik, R. K. Goncharov, and A. D. Kiselev, Asymmetry effects in homodyne and heterodyne measurements: Positive operator-valued measures and asymptotic security of gaussian-continuous-variable quantum key distribution, *Phys. Rev. A* **114**, 012614 (2026).
- [26] M. Navascués, F. Grosshans, and A. Acín, Optimality of gaussian attacks in continuous-variable quantum cryptography, *Physical Review Letters* **97**, 190502 (2006).
- [27] R. García-Patrón and N. J. Cerf, Unconditional optimality of gaussian attacks against continuous-variable quantum key distribution, *Physical Review Letters* **97**, 190503 (2006).
- [28] A. Serafini, *Quantum Continuous Variables: A Primer of Theoretical Methods*, 2nd ed. (CRC Press, Taylor & Francis Group, Boca Raton, 2023) p. 362.
- [29] M. A. Nielsen and I. L. Chuang, *Quantum Computation and Quantum Information*, 10th ed. (Cambridge University Press, NY, 2010) p. 676.
- [30] F. Laudenbach, C. Pacher, C.-H. F. Fung, A. Poppe, M. Peev, B. Schrenk, M. Hentschel, P. Walther, and H. Hübel, Continuous-variable quantum key distribution with Gaussian modulation—the theory of practical implementations, *Advanced Quantum Technologies* **1**, 1800011 (2018).
- [31] I. Devetak and A. Winter, Distillation of secret key and entanglement from quantum states, *Proceedings of the Royal Society A: Mathematical, Physical and Engineering Sciences* **461**, 207 (2005).
- [32] S. M. Barnett, J. Jeffers, A. Gatti, and R. Loudon, Quantum optics of lossy beam splitters, *Physical Review A* **57**, 2134–2145 (1998).
- [33] L. Knöll, S. Scheel, E. Schmidt, D.-G. Welsch, and A. V. Chizhov, Quantum-state transformation by dispersive and absorbing four-port devices, *Physical Review A* **59**, 4716–4726 (1999).
- [34] R. Uppu, T. A. W. Wolterink, T. B. H. Tentrup, and P. W. H. Pinkse, Quantum optics of lossy asymmetric beam splitters, *Optics Express* **24**, 16440 (2016).